

Whitehead's theorem does not say that two CW complexes X and Y with isomorphic homotopy groups are homotopy equivalent, since there is a big difference between saying that X and Y have isomorphic homotopy groups and saying that there is a map $X \rightarrow Y$ inducing isomorphisms on homotopy groups. For example, consider $X = \mathbb{R}P^2$ and $Y = S^2 \times \mathbb{R}P^\infty$. These both have fundamental group $\mathbb{Z}_2$, and Proposition 4.1 implies that their higher homotopy groups are isomorphic since their universal covers S^2 and $S^2 \times S^\infty$ are homotopy equivalent, S^∞ being contractible. But $\mathbb{R}P^2$ and $S^2 \times \mathbb{R}P^\infty$ are not homotopy equivalent since their homology groups are vastly different, $S^2 \times \mathbb{R}P^\infty$ having nonvanishing homology in infinitely many dimensions since it retracts onto $\mathbb{R}P^\infty$. Another pair of CW complexes that are not homotopy equivalent but have isomorphic homotopy groups is S^2 and $S^3 \times \mathbb{C}P^\infty$, as we shall see in Example 4.51.

One very special case when the homotopy type of a CW complex is determined by its homotopy groups is when all the homotopy groups are trivial, for then the inclusion map of a 0-cell into the complex induces an isomorphism on homotopy groups, so the complex deformation retracts to the 0-cell.

Somewhat similar in spirit to the compression lemma is the following rather basic **extension lemma**:

Lemma 4.7. *Given a CW pair (X, A) and a map $f: A \rightarrow Y$ with Y path-connected, then f can be extended to a map $X \rightarrow Y$ if $\pi_{n-1}(Y) = 0$ for all n such that $X - A$ has cells of dimension n .*

Proof: Assume inductively that f has been extended over the $(n-1)$ -skeleton. Then an extension over an n -cell exists iff the composition of the cell's attaching map $S^{n-1} \rightarrow X^{n-1}$ with $f: X^{n-1} \rightarrow Y$ is nullhomotopic. $\square$

Cellular Approximation

When we showed that $\pi_1(S^k) = 0$ for $k > 1$ in Proposition 1.14, we first showed that every loop in S^k can be deformed to miss at least one point if $k > 1$, then we used the fact that the complement of a point in S^k is contractible to finish the proof. The same strategy could be used to show that $\pi_n(S^k) = 0$ for $n < k$ if we could do the first step of deforming a map $S^n \rightarrow S^k$ to be nonsurjective. One might at first think this step was unnecessary, that no continuous map $S^n \rightarrow S^k$ could be surjective when $n < k$, but it is not hard to use space-filling curves from point-set topology to produce such maps. Some work must then be done to construct homotopies eliminating this rather strange behavior.

For maps between CW complexes it turns out to be sufficient for this and many other purposes in homotopy theory to require just that cells map to cells of the same or lower dimension. Such a map $f: X \rightarrow Y$, satisfying $f(X^n) \subset Y^n$ for all n , is called

a **cellular map**. It is a fundamental fact that arbitrary maps can always be deformed to be cellular. This is the **cellular approximation theorem**:

Theorem 4.8. *Every map $f: X \rightarrow Y$ of CW complexes is homotopic to a cellular map. If f is already cellular on a subcomplex $A \subset X$, the homotopy may be taken to be stationary on A .*

Corollary 4.9. $\pi_n(S^k) = 0$ for $n < k$.

Proof: If S^n and S^k are given their usual CW structures, with the 0-cells as basepoints, then every basepoint-preserving map $S^n \rightarrow S^k$ can be homotoped, fixing the basepoint, to be cellular, and hence constant if $n < k$. $\square$

Linear maps cannot increase dimension, so one might try to show that arbitrary maps can be homotoped to maps with some sort of linearity properties. One of the oldest results of this sort is the simplicial approximation theorem in §2.C. Cellular approximation can be regarded as an analog for CW complexes of simplicial approximation for simplicial complexes since simplicial maps are cellular. However, simplicial maps are much more rigid than cellular maps, which perhaps explains why subdivision of the domain is required for simplicial approximation but not for cellular approximation. The core of the proof of cellular approximation will be a weak form of simplicial approximation that can be proved by a rather elementary direct argument.

Proof of 4.8: Suppose inductively that $f: X \rightarrow Y$ is already cellular on the skeleton X^{n-1} , and let e^n be an n -cell of X . The closure of e^n in X is compact, being the image of a characteristic map for e^n , so f takes the closure of e^n to a compact set in Y . Since a compact set in a CW complex can meet only finitely many cells by Proposition A.1 in the Appendix, it follows that $f(e^n)$ meets only finitely many cells of Y . Let $e^k \subset Y$ be a cell of highest dimension meeting $f(e^n)$. We may assume $k > n$, otherwise f is already cellular on e^n . We will show below that it is possible to deform $f|_{X^{n-1} \cup e^n}$, staying fixed on X^{n-1} , so that $f(e^n)$ misses some point $p \in e^k$. Then we can deform $f|_{X^{n-1} \cup e^n} \text{ rel } X^{n-1}$ so that $f(e^n)$ misses the whole cell e^k by composing with a deformation retraction of $Y^k - \{p\}$ onto $Y^k - e^k$. By finitely many iterations of this process we eventually make $f(e^n)$ miss all cells of dimension greater than n . Doing this for all n -cells, staying fixed on n -cells in A where f is already cellular, we obtain a homotopy of $f|_{X^n} \text{ rel } X^{n-1} \cup A^n$ to a cellular map. The induction step is then completed by appealing to the homotopy extension property in Proposition 0.16 to extend this homotopy, together with the constant homotopy on A , to a homotopy defined on all of X . Letting n go to ∞ , the resulting possibly infinite string of homotopies can be realized as a single homotopy by performing the n^{th} homotopy during the t -interval $[1 - 1/2^n, 1 - 1/2^{n+1}]$. This makes sense since each point of X lies in some X^n , which is eventually stationary in the infinite chain of homotopies.

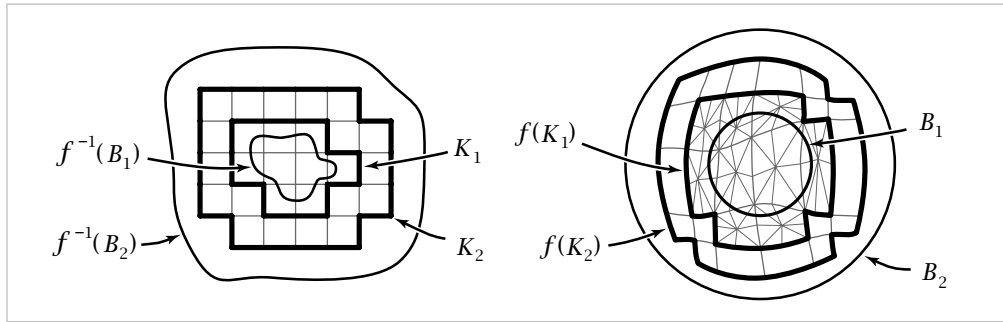
To fill in the missing step in this argument we will use the following technical statement:

Lemma 4.10. *Let $f: I^n \rightarrow Z$ be a map, where Z is obtained from a subspace W by attaching a cell e^k . Then f is homotopic rel $f^{-1}(W)$ to a map f_1 for which there is a simplex $\Delta^k \subset e^k$ with $f_1^{-1}(\Delta^k)$ a union (possibly empty) of finitely many convex polyhedra, on each of which f_1 is the restriction of a linear surjection $\mathbb{R}^n \rightarrow \mathbb{R}^k$.*

Here a convex polyhedron in $I^n \subset \mathbb{R}^n$ is any subspace that can be obtained as the intersection of a finite number of half-spaces defined by linear inequalities of the form $\sum_i a_i x_i \leq b$.

Before proving the lemma, let us see how it finishes the proof of the cellular approximation theorem. Composing the given map $f: X^{n-1} \cup e^n \rightarrow Y^k$ with a characteristic map $I^n \rightarrow X$ for e^n , we obtain a map f as in the lemma, with $Z = Y^k$ and $W = Y^k - e^k$. The homotopy given by the lemma is fixed on ∂I^n , hence induces a homotopy f_t of $f|_{X^{n-1} \cup e^n}$ fixed on X^{n-1} . If $k > n$, there are no surjective linear maps $\mathbb{R}^n \rightarrow \mathbb{R}^k$, so $f_1^{-1}(\Delta^k)$ must be empty, and we can choose p to be any point of Δ^k . $\square$

Proof of 4.10: Identifying e^k with $\mathbb{R}^k$, let $B_1, B_2 \subset e^k$ be the closed balls of radius 1 and 2 centered at the origin. Since $f^{-1}(B_2)$ is closed and therefore compact in I^n , it follows that f is uniformly continuous on $f^{-1}(B_2)$. Thus there exists $\varepsilon > 0$ such that $|x - y| < \varepsilon$ implies $|f(x) - f(y)| < 1/2$ for all $x, y \in f^{-1}(B_2)$. Subdivide the interval I so that the induced subdivision of I^n into cubes has each cube lying in a ball of diameter less than ε . Let K_1 be the union of all the closed cubes meeting $f^{-1}(B_1)$, and let K_2 be the union of all the closed cubes meeting K_1 . Then we have inclusions $f^{-1}(B_1) \subset K_1 \subset K_2 \subset f^{-1}(B_2)$, the last one because points of $f(K_2)$ have distance less than $1/2$ from $f(K_1)$ and points of $f(K_1)$ have distance less than $1/2$ from B_1 .



We can view K_2 as a CW complex whose i -cells are i -dimensional open cubes, the interiors of the i -dimensional faces of the k -dimensional cubes of K_2 for $i \leq k$. The barycentric subdivision of this cubical cell structure is a simplicial complex structure on K_2 whose vertices are the center points of the cells. One can build this simplicial structure inductively over skeleta of the cubical cell structure, the induction step being

to cone off the simplicial structure on the boundary of each cubical cell to the center point of the cell.

Let $g: K_2 \rightarrow e^k = \mathbb{R}^k$ be the map that equals f on all vertices of simplices of the subdivision and is linear on each simplex. Define a homotopy $f_t: K_2 \rightarrow e^k$ by the formula $(1 - t\varphi)f + (t\varphi)g$ where $\varphi: K_2 \rightarrow [0, 1]$ is a map with $\varphi(\partial K_2) = 0$ and $\varphi(K_1) = 1$. Thus $f_0 = f$ and $f_1|_{K_1} = g|_{K_1}$. Since f_t is the constant homotopy on ∂K_2 , we may extend f_t to be the constant homotopy of f on the rest of I^n .

We claim that there is a neighborhood N of 0 in B_1 such that $f_1^{-1}(N) \subset K_1$. This is equivalent to saying that f_1 sends the complement of K_1 to the complement of N . Points in the complement of K_2 are no problem since $f_1 = f$ on such points and f sends the complement of K_2 to the complement of B_1 . For points of $K_2 - K_1$ consider a simplex σ of the subdivision of K_2 . This is mapped by f into a ball B_σ of radius $1/2$. Since B_σ is convex, g also maps σ into B_σ , and therefore so does f_1 . If σ is not contained in K_1 , then B_σ meets the exterior of B_1 and hence is disjoint from a neighborhood of 0 in B_1 . Since there are only finitely many σ 's, there is a neighborhood N of 0 in B_1 disjoint from $f_1(\sigma)$ for all σ not contained in K_1 . For this N we have $f_1^{-1}(N) \subset K_1$.

For a simplex $\Delta^k \subset N$, the preimage $f_1^{-1}(\Delta^k) \subset K_1$ is the union of its intersections with simplices σ of K_1 , and each such intersection is a convex polyhedron since it is the intersection of σ with the convex polyhedron $L_\sigma^{-1}(\Delta^k)$ where $L_\sigma: \mathbb{R}^n \rightarrow \mathbb{R}^k$ is the linear map restricting to g on σ . (Recall that $f_1 = g$ on K_1 .) To finish the proof it therefore suffices to choose Δ^k to be disjoint from the images of all the nonsurjective L_σ 's, which is certainly possible since these images consist of finitely many hyperplanes of dimension less than k . $\square$

Example 4.11: Cellular Approximation for Pairs. Every map $f: (X, A) \rightarrow (Y, B)$ of CW pairs can be deformed through maps $(X, A) \rightarrow (Y, B)$ to a cellular map. This follows from the theorem by first deforming the restriction $f: A \rightarrow B$ to be cellular, then extending this to a homotopy of f on all of X , then deforming the resulting map to be cellular staying fixed on A . As a further refinement, the homotopy of f can be taken to be stationary on any subcomplex of X where f is already cellular.

An easy consequence of this is:

Corollary 4.12. *A CW pair (X, A) is n -connected if all the cells in $X - A$ have dimension greater than n . In particular the pair (X, X^n) is n -connected, hence the inclusion $X^n \hookrightarrow X$ induces isomorphisms on π_i for $i < n$ and a surjection on π_n .*

Proof: Applying cellular approximation to maps $(D^i, \partial D^i) \rightarrow (X, A)$ with $i \leq n$ gives the first statement. The last statement comes from the long exact sequence of the pair (X, X^n) . $\square$

CW Approximation

A map $f: X \rightarrow Y$ is called a **weak homotopy equivalence** if it induces isomorphisms $\pi_n(X, x_0) \rightarrow \pi_n(Y, f(x_0))$ for all $n \geq 0$ and all choices of basepoint x_0 . Whitehead's theorem can be restated as saying that a weak homotopy equivalence between CW complexes is a homotopy equivalence. It follows easily that this holds also for spaces homotopy equivalent to CW complexes. In general, however, weak homotopy equivalence is strictly weaker than homotopy equivalence. For example, there exist noncontractible spaces whose homotopy groups are all trivial, such as the 'quasi-circle' according to an exercise at the end of this section, and for such spaces a map to a point is a weak homotopy equivalence that is not a homotopy equivalence.

One of the more important results in this subsection is that for every space X there is a CW complex Z and a weak homotopy equivalence $f: Z \rightarrow X$. Such a map $f: Z \rightarrow X$ is called a **CW approximation** to X . A weak homotopy equivalence induces isomorphisms on all homology and cohomology groups, as we will show, so CW approximations allow many general statements in algebraic topology to be reduced to the case of CW complexes, where one can often make cell-by-cell arguments.

The technique for constructing CW approximations can be used to do other things as well. For a start, one could try for a relative version in which X is assumed to contain a subspace A which is already a CW complex and Z is constructed to contain A as a subcomplex, with $f: Z \rightarrow X$ restricting to the identity map on A . Next, if (X, A) is n -connected, one could try to make (Z, A) n -connected in the strong geometric sense that all cells of $Z - A$ have dimension greater than n . In fact, it turns out to be possible to do a construction satisfying the latter condition even if (X, A) is not n -connected, but in this case $\pi_i(Z)$ for $i < n$ will be isomorphic to $\pi_i(A)$ rather than to $\pi_i(X)$.

Here is a definition that is sufficiently general to cover all these situations. Given a pair (X, A) where the subspace $A \subset X$ is a nonempty CW complex, an **n -connected CW model for (X, A)** is an n -connected CW pair (Z, A) and a map $f: Z \rightarrow X$ with $f|_A$ the identity, such that $f_*: \pi_i(Z) \rightarrow \pi_i(X)$ is an isomorphism for $i > n$ and an injection for $i = n$, for all choices of basepoint. Since (Z, A) is n -connected, the map $\pi_i(A) \rightarrow \pi_i(Z)$ is an isomorphism for $i < n$ and a surjection for $i = n$. In the critical dimension n , the maps $A \hookrightarrow Z \xrightarrow{f} X$ induce a composition $\pi_n(A) \rightarrow \pi_n(Z) \rightarrow \pi_n(X)$ factoring the map $\pi_n(A) \rightarrow \pi_n(X)$ as a surjection followed by an injection, just as any homomorphism $\varphi: G \rightarrow H$ can be factored (uniquely) as a surjection $\varphi: G \rightarrow \text{Im } \varphi$ followed by an injection $\text{Im } \varphi \hookrightarrow H$. One can think of Z as a sort of homotopy-theoretic hybrid of A and X . As n increases, the hybrid looks more and more like A , and less and less like X .

This definition specializes to the earlier notion of a CW approximation by taking $n = 0$ and letting A consist of one point in each path-component of X . This forces $f_*: \pi_0(Z) \rightarrow \pi_0(X)$ to be surjective as well as injective.